DO NOW — RECALL

Lesson 071 · Pipelining and Computational Methods Rev

Name: _____ Class: _____ Date: _____

1. What is a heuristic?

2. When does backtracking undo a choice?

3. What does throughput measure?

4. Rising-edge D flip-flop Q=1, D=0 immediately before active edge. What is Q after the edge?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Estimate guiding choices, not necessarily guaranteeing optimum.

2. Model answer

When the branch cannot produce a valid result.

3. Model answer

Completed items per unit time.

4. Model answer

Q becomes 0; D is sampled at the active clock edge.

By the end, you can…

• Pipelining overlaps separate stages of different tasks; it increases throughput after filling the pipeline, not necessarily single-task latency.

• Four tasks on a three-stage pipeline complete in six ideal cycles, versus twelve when each task runs all stages in isolation.

• Choose the computational method that fits: decomposition for parts; backtracking for reversible choices; heuristic for guided estimates; visualisation for patterns.

Vocabulary — know these terms

pipeline — Overlapping multiple processing stages.

throughput — Work completed per time.

latency — Time for one task to finish.

stall — Delay from dependency or resource issue.

NEW LEARNING

Read, annotate and explain

Observe

Pipelining overlaps separate stages of different tasks; it increases throughput after filling the pipeline, not necessarily single-task latency.

Explain

Four tasks on a three-stage pipeline complete in six ideal cycles, versus twelve when each task runs all stages in isolation.

Evaluate

Choose the computational method that fits: decomposition for parts; backtracking for reversible choices; heuristic for guided estimates; visualisation for patterns.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Find ideal cycles for 3 tasks through 2 single-cycle stages.

Explain

Start a new task each cycle while earlier tasks advance; last task finishes cycle4.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Draw ideal 2-stage pipeline schedule for 3 tasks and state completion cycle. [6]

Question 2

For 6 tasks and 4 one-cycle stages, compare pipelined and sequential ideal times. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

Match backtracking, data mining, visualisation and pipelining to four problems: maze with dead ends, shopping patterns, charting results and overlapping video processing. Explain each. [8]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

A pipeline always divides each individual task's latency by its number of stages.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.